Amutheezan Sivagnanam

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SUMMARY

— **PhD Candidate** and **Machine Learning Researcher** specializing in **Deep Reinforcement Learning**
— **Six** years of academic research experience building **scalable reinforcement learning** and **deep learning** systems for efficient ML **training** and **deployment**
— **Two** years of industrial experience in **software development**, **behavioral driven testing** and **CI/CD** while following **agile practices**
— Good **publication** record with **an h-index of 6 and 100+ citations**, featuring in top-tier **AI and ML** conferences
— **Work Authorization Status: U.S. Permanent Resident (Green Card holder) | Open to relocation**

EDUCATION

Ph.D. Computer Science, (August 2025): **Pennsylvania State University**, University Park, PA, Advisor: Dr. Aron Laszka
- Dissertation: Application of Deep Reinforcement Learning to Solve Optimization Problems in Transportation Domains
M.S. Computer Science, (August 2022): **University of Houston**, Houston, TX, Advisor: Dr. Aron Laszka

RESEARCH EXPERIENCE

Graduate Research Assistant, *Pennsylvania State University*, (August 2022 - May 2025)
— Devised and evaluated a **Transformer** based **multi-agent reinforcement learning algorithm** to solve the **emergency responder reallocation problem** that **reduces** decision making time **by 3 orders of magnitude**
— Leveraged a **deep reinforcement learning** approach to tackle the challenge of online vehicle routing with advance booking, enabling **prompt confirmation within seconds**
Graduate Research Assistant, *University of Houston*, (September 2019 - August 2022)
— Introduced a novel **reinforcement learning** based solution approach to solve the problem of **online booking for offline vehicle routing problem** which resulted in **20 % reduction of operational costs** for the transit agency
— Introduced **novel problem formulation and algorithms** that l**ower the energy costs by $140k** annually for public transit agencies operating mixed fleets of buses
— **Skills**: **Python**, C++, Java, C, SQL, MySQL, MongoDB, SQLite, **PyTorch**, TensorFlow, Scikit-learn, Transformers, Machine Learning, Deep Learning, Computer Vision, **Reinforcement Learning**, DQN, DDPG, TRPO, PPO, GRPO, LLM, RLHF, RAG, Prompt Engineering, LangChain, Bayesian Optimization, Data Science, Data Visualization, Clustering, Statistical Analysis, Data Mining, Ray, RLLib, OpenAI Gym, NumPy, pandas, matplotlib, Linear Programming, Integer Programming, CPLEX, Gurobi, Mosek, OR-Tools, Spark, **Amazon Web Services (Sage Maker, S3, Lambda)**, Git, Docker, Linux, Bash, Object-Oriented Development, MATLAB

SELECT PUBLICATIONS

A. Sivagnanam et al.; Multi-Agent Reinforcement Learning with Hierarchical Coordination for Emergency Responder Stationing. International Conference on Machine Learning. **ICML** 2024. **Core Ranking A***
A. Sivagnanam et al.; Offline Vehicle Routing Problem with Online Bookings: A Novel Problem Formulation with Applications to Paratransit. International Joint Conference on Artificial Intelligence. **IJCAI** 2022. **Core Ranking A***, Invited Talk: Vienna Austria
A. Sivagnanam et al.; Minimizing Energy Use of Mixed-Fleet Public Transit for Fixed-Route Service. AAAI Conference on Artificial Intelligence. **AAAI** 2021. **Core Ranking A***, Invited Talk: Virtual Conference

SOFTWARE ENGINEERING EXPERIENCE

Software Engineer, *LSEG Technology*, (January 2018 - July 2019)
— Introduced **unit testing** for libraries in the Post-Trade **C++** codebase. Implemented and validated **database changes** for Post-Trade products for the Singapore Stock Exchange using **Behavior Driven Development** (BDD) testing approaches in **Java**. Contributed to the **CI/CD pipeline** of the Post-Trade product using **Python and Git**.
Software Engineering Intern, *WSO2 Lanka PVT Ltd*, (July 2016 - December 2016)
— Engineered an **alert generation system** to monitor disease outbreaks by analyzing descriptive **HL7/FHIR data**, automating email and SMS notifications, and supporting rapid decision-making through prompt data processing. Developed a mechanism to evaluate hospital functionality by assessing bed and oxygen cylinder availability from admission and discharge messages, demonstrating **effective requirements gathering** and integration of **structured data processing.**
— **Skills**: Python, C++, JAVA, JS, PHP, SQL, Object Oriented Development, Github, Linux, Agile Practices, Behaviour Driven Development, **Natural Language Processing**, Data Science, Data Visualization, **HL7**, **FHIR**, **CI/CD**

REFERENCES

Dr. Aron Laszka (PhD Supervisor): aql5923@psu.edu
Assistant Professor, Pennsylvania State University
https://aronlaszka.com

Dr. Abhishek Dubey: abhishek.dubey@vanderbilt.edu
Associate Professor, Vanderbilt University
https://abhishekdubey.bio

Dr. Weidong Shi: wshi3@uh.edu
Associate Professor, University of Houston